THE COLLISION TRANSFORM AND THE CRITICAL STRIP

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Abstract. The centered collision sum $F^\circ(s) = \sum_p S_p^\circ/p^s$ converges at $s = 1$ (proved in the preceding paper). This paper investigates whether convergence extends below $s = 1$ into the critical strip.

The collision transform writes $F^\circ(s)$ as a finite linear combination of character sums $P(s,\chi) = \sum_p \chi(p)/p^s$, linked to $\log L(s,\chi)$ for L-functions modulo $b^{\ell+1}$. Antisymmetry restricts the sum to odd characters. Conditional on a zero-free hypothesis for the relevant L-functions, $F^\circ(s)$ converges for real $s > \sigma_0$, where σ_0 depends on the zero-free region.

Computation suggests convergence persists to $s \approx 0.5$ in base 3 and to $s \approx 0.6$ in bases 5, 7, 10, 12. These observations are consistent with, but do not prove, convergence in the strip.

1. Introduction

This paper investigates the collision transform below $s = 1$.

The preceding paper [7] proved that the centered collision fluctuation sum

$$F^\circ(1,\ell) = \sum_{\chi\neq\chi_0} \hat{S}_\ell(\chi) \sum_p \frac{\chi(p)}{p}$$

converges at $s = 1$. The proof used the finite determination theorem (the collision deviation $S_\ell(p)$ depends only on $p \bmod b^{\ell+1}$), the collision transform (decomposition over Dirichlet characters modulo $b^{\ell+1}$), and Mertens' theorem for arithmetic progressions.

The present paper investigates whether convergence extends below $s = 1$.

Date: January 2024 (revised April 2026).

2020 *Mathematics Subject Classification.* 11A63, 11M06, 11M26.

2. The Character Decomposition Below $s = 1$

For $\mathrm{Re}(s) > 1$, the prime character sum [1, 2]

$$P(s, \chi) = \sum_p \frac{\chi(p)}{p^s}$$

is related to the Dirichlet L-function by

$$P(s, \chi) = \log L(s, \chi) - H(s, \chi), \tag{1}$$

where the higher-power sum $H(s, \chi) = \sum_p \sum_{k \geq 2} \chi(p)^k / (k\, p^{ks})$ converges absolutely for $\mathrm{Re}(s) > 1/2$.

The centered fluctuation sum is therefore

$$F^\circ(s) = -\sum_{\chi \neq \chi_0} \hat{S}(\chi)\, \log L(s, \chi) + \sum_{\chi \neq \chi_0} \hat{S}(\chi)\, H(s, \chi).$$

The second sum converges for $\mathrm{Re}(s) > 1/2$. The convergence of $F^\circ(s)$ at $s = \sigma$ is therefore controlled by the first sum: the weighted combination of $\log L(s, \chi)$ values at $s = \sigma$.

If $L(\sigma, \chi)$ has a zero at some ρ with $\mathrm{Re}(\rho) > \sigma$, then $\log L(s, \chi)$ has a logarithmic singularity at ρ. Convergence of $F^\circ(\sigma)$ despite such singularities would require cancellation across characters: the contributions from zeros of different L-functions, weighted by $\hat{S}(\chi)$, must cancel.

3. Computational Evidence

s	base 3	base 5	base 7	base 10	base 12
1.0	−0.151	−0.187	−0.137	−0.199	−0.278
0.9	−0.168	−0.230	−0.186	−0.282	−0.417
0.8	−0.186	−0.277	−0.250	−0.402	−0.624
0.7	−0.205	−0.322	−0.337	−0.577	−0.926
0.6	−0.227	−0.347	−0.457	−0.835	−1.346
0.5	−0.257	−0.312	−0.636	−1.219	−1.880

Table 1. $F^\circ(s)$ at lag $\ell = 1$ across bases, using 18,000–78,000 primes per base.

The data reveals a base-dependent penetration pattern:

- **Base** 3: drift < 0.003 at all s down to $s = 0.5$. The centered sum appears to converge on the critical line.
- **Base** 5: stable to $s = 0.7$. Drift grows at $s = 0.6$ and $s = 0.5$.
- **Bases** 7, 10: stable to $s = 0.7$. Moderate drift at $s = 0.6$. Growing drift at $s = 0.5$.

s	base 3	base 5	base 7	base 10	base 12
1.0	0.000	0.000	0.000	0.000	0.000
0.9	0.000	0.001	0.001	0.000	0.001
0.8	0.000	0.004	0.002	0.001	0.005
0.7	0.000	0.013	0.007	0.004	0.017
0.6	0.001	0.042	0.027	0.016	0.058
0.5	0.003	0.135	0.099	0.065	0.205

TABLE 2. Drift: absolute change in $F^\circ(s)$ between the first and second halves of the prime range. Small drift indicates convergence. Drift less than 0.01 is consistent with stability.

- **Base** 12: stable to $s = 0.7$. Significant drift at $s = 0.6$ and $s = 0.5$.

4. The Penetration Depth

Definition 1. The *penetration depth* of the collision transform at base b and lag ℓ is

$$\sigma_0(b, \ell) = \inf\{\sigma > 0 : F^\circ(\sigma, \ell) \text{ converges}\}.$$

The data suggests $\sigma_0(3, 1) \leq 0.5$ and $\sigma_0(b, 1) \leq 0.7$ for all bases tested.

The base dependence of the apparent penetration depth may reflect the number of primes required rather than a genuine difference in σ_0. Base 3 has $\phi(9) = 6$ characters modulo $b^2 = 9$, while base 12 has $\phi(144) = 48$. Cancellation among more characters requires more primes to manifest.

Conjecture 2 (Universal penetration)**.** *For every base $b \geq 2$ and lag $\ell \geq 1$: $\sigma_0(b, \ell) = 1/2$.*

This conjecture is consistent with GRH. By equation (1), the convergence of $F^\circ(s)$ for $\mathrm{Re}(s) > 1/2$ is equivalent to the convergence of $\sum_{\chi \neq \chi_0} \hat{S}(\chi) \log L(s, \chi)$ for $\mathrm{Re}(s) > 1/2$. Under GRH, the individual $P(s, \chi)$ converge for $\mathrm{Re}(s) > 1/2$, and the finite sum $F^\circ(s)$ inherits this convergence.

5. The Antisymmetry Theorem

Theorem 3 (Odd-character restriction)**.** *The centered collision transform satisfies $\hat{S}^\circ(\chi) = 0$ for every even Dirichlet character χ modulo $b^{\ell+1}$.*

Proof. By the reflection identity [7], $S(a)+S(m-a)=-1$ for $m=b^{\ell+1}$. The map $a\mapsto m-a$ sends spectral class R to class $b-2-R$, with class means satisfying $\overline{S}_R+\overline{S}_{b-2-R}=-1$. Therefore $S^\circ(a)+S^\circ(m-a)=0$.

For any character χ, $\chi(m-a)=\chi(-1)\chi(a)$. If χ is even ($\chi(-1)=1$):

$$\hat{S}^\circ(\chi)=\frac{1}{\phi(m)}\sum_a S^\circ(a)\,\overline{\chi}(a)=\frac{1}{\phi(m)}\sum_a S^\circ(a)\,\overline{\chi}(m-a)=-\hat{S}^\circ(\chi),$$

using $S^\circ(m-a)=-S^\circ(a)$ and $\overline{\chi}(m-a)=\overline{\chi}(a)$. Therefore $\hat{S}^\circ(\chi)=0$. □

Corollary 4. *The centered fluctuation sum decomposes as*

$$F^\circ(s)=\sum_{\substack{\chi \bmod m\\ \chi(-1)=-1}}\hat{S}^\circ(\chi)\,P(s,\chi).$$

Only odd L-functions control the penetration depth.

6. Positivity of Character Products

At the small bases displayed below, each odd character χ contributing to F° has

$$\mathrm{Re}\big(\hat{S}^\circ(\chi)\,P(1/2,\chi)\big)\geq 0.$$

This is only a finite observation, not a general positivity law.

base	odd χ	$\lvert\hat{S}^\circ\rvert$	$\lvert P(1/2)\rvert$	$\mathrm{Re}(\hat{S}^\circ\cdot P)$
3	$j=1$	0.667	0.469	+0.309
3	$j=5$	0.667	0.469	+0.309
5	$j=3$	0.828	0.520	+0.368
5	$j=17$	0.828	0.520	+0.368

Table 3. Character products at selected small bases. All displayed real parts are non-negative. Later computation at base 10 shows that mixed signs occur in general.

Later computation shows positivity fails in general: at base 10, 7 of 20 products have negative real part. Convergence of $F^\circ(s)$ below $s=1$ therefore depends on cancellation between terms, not on individual term convergence.

Remark 5 (Conditional penetration). The following conditional statement illustrates one possible route from the collision transform to GRH, though the required positivity hypothesis has since been shown to fail in general.

If the products $\hat{S}^\circ(\chi) \cdot P(s,\chi)$ had non-negative real part for all odd χ and all $s \geq 1/2$, then:

$$F^\circ(1/2) \text{ converges } \iff L(s,\chi) \neq 0 \text{ for } \operatorname{Re}(s) > \tfrac{1}{2},\ \chi \in \mathcal{X}^-.$$

Since positivity fails, the reverse direction of this argument does not apply. Convergence of $F^\circ(s)$ below $s = 1$ may occur through cancellation between terms of mixed sign, without requiring each individual L-function to be zero-free. The relationship between F° and GRH remains open.

7. The Aperiodic Residual

The finite determination theorem shows that $S_\ell(p)$ depends only on $p \bmod b^{\ell+1}$. The centered version S° removes the class mean. A deeper question: does the residual S° carry additional modular structure at higher resolution?

In the tested lag-1 cases, computation shows it does not. For each lag-1 class $a \bmod b^2$, the sub-classes $a' \bmod b^3$ satisfying $a' \equiv a \pmod{b^2}$ all have identical mean S values. For these examples, the lag-1 residual carries no additional modular information at lag 2.

base	lag-1 mod	lag-2 mod	sub-classes	residual
3	9	27	3 per class	0 everywhere
5	25	125	5 per class	0 everywhere

Table 4. Tested lag-1 towers in bases 3 and 5. In these examples, the lag-1 residual does not split further at lag 2.

These examples suggest that the centered deviation S° behaves aperiodically after the lag-1 class mean is removed: in the tested towers, it is not refined by passing from b^2 to b^3. The stamp at each lag may capture all of the modular structure visible at that lag, with the remainder behaving as a non-periodic fluctuation.

The convergence of $F^\circ(s)$ at $s = \sigma$ is therefore, at least heuristically, a statement about whether this residual cancels at weighting $1/p^\sigma$. At $s = 1$, the centered sum cancels by the collision transform (proved). Whether convergence extends below $s = 1$ depends on cancellation between terms of mixed sign and remains open.

8. The Base Sweep

The preceding sections analyze the collision transform at a single base. A different computational perspective emerges from sweeping

across a finite family of bases. For a fixed prime p and a finite tested set $\mathcal{B}$, define the base-swept resistance

$$\mathcal{R}_{\mathcal{B},w}(p) = \sum_{b\in\mathcal{B}} w(b)\, S_1(p,b),$$

where $w(b)$ is a chosen weight on $\mathcal{B}$. This aggregates the collision deviation of a single prime across the tested bases simultaneously.

In the tested sweeps, $\mathcal{R}_{\mathcal{B},w}(p)$ splits by $p \bmod 3$: primes $\equiv 1 \pmod 3$ carry approximately $2/3$ of the total resistance and primes $\equiv 2 \pmod 3$ carry approximately $1/3$. This ratio is stable across the tested weights $w = 1/b$, $w = 1$, $w = 1/\sqrt{b}$, and $w = 1/b^2$, but no general invariance theorem is claimed here.

The mod-3 structure is not visible from any single base. In these experiments, it emerges only from the collective instrument.

9. The Mod-3 Polarity

In the tested cases, the centered residual S° shows no modular structure in the base direction: the lag tower appears flat (Section 4). But it has structure in the *prime* direction. Its mean over primes $p \equiv 1 \pmod 3$ differs from its mean over primes $p \equiv 2 \pmod 3$. In base 10 at lag 1, the centered values $S^\circ(a)$ for units $a \equiv 1 \pmod 3$ average -1.30, while those for $a \equiv 2 \pmod 3$ average 0.

In the tested base sweeps, this asymmetry is stable under the reweightings described in the previous section. A base-swept resistance

$$\textstyle\sum_{b\in\mathcal{B}} w(b)\, S_1(p,b)$$

splits by $p \bmod 3$: primes $\equiv 1 \pmod 3$ carry approximately $2/3$ of the total resistance and primes $\equiv 2 \pmod 3$ carry approximately $1/3$.

Despite this asymmetry, the antisymmetry $S^\circ(a) = -S^\circ(m-a)$ is compatible with the mod-3 structure because the reflection $a \mapsto m-a$ respects the mod-3 pairing (in base 10: $m = 100 \equiv 1 \pmod 3$).

However, in the computations reported here, removing the mod-3 bias (doubly centering by both spectral class and $p \bmod 3$) destroys the apparent penetration depth. The doubly centered sum $F^{\circ\circ}$ still exhibits the same antisymmetry, but in the tested data it grows rapidly for $s < 1$.

The data suggests that the mod-3 structure is not needed for convergence at $s = 1$, but may matter for any deeper penetration below $s = 1$. At present this is an observed feature of the computations, not a theorem.

The fraction $2/3$ that governs this distribution is the alignment limit of the prime 3 from the first paper in this series [4]: the constant $(p-1)/p$

s	F°	$F^{\circ\circ}$
1.0	0.08	0.77
0.7	0.27	12.3
0.5	0.46	111

TABLE 5. Removing the mod-3 component in the tested base-3 data leaves the $s = 1$ value bounded but produces rapid growth below $s = 1$. This suggests that the mod-3 polarity is relevant to penetration depth.

evaluated at the prime the golden ratio selects. Its reappearance here, as the observed fraction of base-swept resistance carried by primes $\equiv 1 \pmod 3$, is suggestive: the constant that began the program as a threshold for digit matching may also be connected to penetration into the critical strip.

10. Remarks

The instrument. The collision transform was introduced in [7] to prove the Centered Collision PNT at $s = 1$. The present paper observes that the same instrument sees deeper. The digit function $\delta(r) = \lfloor br/p \rfloor$, through its collision statistics, produces a Dirichlet series whose convergence appears, in the tested data, to extend into the critical strip. The depth of this apparent region is a new computational invariant of the collision structure.

Base as instrument. The conventional view treats the positional base as a notational choice with no mathematical content. This program suggests otherwise. The base determines the bin partition and the number of characters in the collision transform, and at finite range the observed penetration depth varies with the base. Different bases are different instruments: base 3 has six characters and, in the present data, reaches $s = 0.5$; base 12 has forty-eight and reaches only $s = 0.7$ at the same computational depth. The base is not mere notation here. It acts as the resolution of the instrument that probes the prime.

The equivalence. The digit function $\delta(r) = \lfloor br/p \rfloor$ encodes the prime's internal geometry. The collision transform reads a finite stamp (the modular structure) and isolates the aperiodic residual. The antisymmetry restricts the residual to odd characters. The character products have mixed signs, so convergence below $s = 1$ depends on cancellation between terms. Whether this cancellation extends to $s = 1/2$, and how it relates to the zeros of the relevant L-functions, remains open.

The arc. The program began with the alignment of the prime 3 approaching 2/3 [4]. It built through the gate width theorem [5], the collision transform [7], and arrives here at the critical strip. Each step revealed structure or a new question: the gate width is universal, the collision deviation is finitely determined, the centered residual appears aperiodic in the tested towers, and the problem of convergence at $s = 1/2$ brings the program into direct contact with the zero distribution of the relevant family of L-functions. The inside of the prime and the zeros of the L-function are connected through the collision transform, but the full relationship remains open.

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